Available online at www.sciencedirect.com
SCIENCE DIRECT®
Physics of the Earth and Planetary Interiors 139 (2003) 285–303
PHYSICS OF THE EARTH AND PLANETARY INTERIORS
www.elsevier.com/locate/pepi

Absolute paleointensities and paleodirections of miocene and pliocene lavas from Fernando de Noronha, Brazil

R. Leonhardt [a,*], J. Matzka [a], E.A. Menor [b]

[a] *Department for Earth and Environmental Sciences, Geophysics Section, Ludwig-Maximilians-Universität, München, Germany*
[b] *Engenharia de Minas, Universidade Federal de Pernambuco, Recife, Pernambuco, Brazil*

Received 22 June 2003; received in revised form 8 September 2003; accepted 9 September 2003

Abstract

Forty-four volcanic units were sampled on the Equatorial Atlantic Island of Fernando de Noronha, Brazil. Twenty-seven sites are from the 1.8–3.3 Myr old Quixaba formation, and the others are from the approximately 8–12 Myr old Remedios formation. Analyses of paleodirections yielded normal and reverse polarities in both formations with virtual geomagnetic poles (VGPs) and VGP dispersion (S_f) of VGP_{lat}: 86.9°, VGP_{long}: 136.3°, S_f: 11.2° for the Quixaba formation and VGP_{lat}: 86.8°, VGP_{long}: 196.7°, S_f: 15.5° for the Remedios formation. Successful paleointensity determinations were obtained for 9 of the 27 sampled units in the Quixaba formation. A modified Thellier method was used that includes checks to monitor alterations and repetition of demagnetization steps to detect multidomain tails of partial thermoremanent magnetizations (pTRM) and magnetization stabilization processes. Furthermore, the additivity of pTRMs was monitored during this modified technique of the Thellier experiment. These additivity checks provide a quality test for correcting the paleointensity determination for magnetomineralogical changes which occur during laboratory treatment. Such correction methods were used for analyzing approximately 50% of the determinations and lead to a reduction of the within-site scatter. The mean paleointensity result of 29.3 μT, corresponding to a virtual axial dipole moment (VADM) of 7.5×10^{22} A m^2, yields approximately the present day field strength. Mean values of the paleodirections for volcanic units of reversed and normal polarity are closer to a dipolar field configuration than previously observed. Today's geomagnetic field in the Equatorial Atlantic shows a significant dipole anomaly. Our results indicate that this anomaly was far less pronounced in the geological past.
© 2003 Elsevier B.V. All rights reserved.

Keywords: Paleomagnetism; Absolute paleointensity; Geomagnetic field; Fernando de Noronha, Brazil

1. Introduction

Paleomagnetic results of Tertiary lava flows are important aspects in the study of long-term average features of the Earth's magnetic field. One of the basic assumptions of paleomagnetism is that the geomagnetic field is on average an axial dipole, parallel to the Earth's axis of rotation. However, paleomagnetic data of the past million years show persistent and systematic departures from this simple model (Gubbins and Kelly, 1993). McElhinny et al. (1996) found that a purely zonal field geometry of an axial dipole and an axial quadrupole is sufficient to fit all available paleomagnetic data for the past 5 Myr from lava flows and sediments. Johnson and Constable (1995, 1997), however, require non-zonal terms in the spherical

* Corresponding author. Tel.: +49-89-2180-4231; fax: +49-89-2394-4205.
E-mail address: leon@geophysik.uni-muenchen.de (R. Leonhardt).

0031-9201/$ – see front matter © 2003 Elsevier B.V. All rights reserved.
doi:10.1016/j.pepi.2003.09.008

harmonic description of the time-averaged paleomagnetic field using lava flow data. In particular Johnson and Constable (1995) describe a persistent hemispheric asymmetry and an asymmetry between the normal and the reversed geomagnetic field in their model, which later was attributed to be an artifact due to poor distribution of and unrecognized errors within the data (Johnson and Constable, 1997; Carlut and Courtillot, 1998). Paleointensity data has been used for modeling (e.g. Kono and Hiroi, 1996), but the paleointensity data set is, as yet, insufficient to obtain detailed models of time-averaged paleomagnetic field (e.g. Hatakeyama and Kono, 2002). Furthermore, absolute paleointensity data for the last 5 Myr is characterized by large fluctuations and uncertainties (Juarez and Tauxe, 2000).

A pronounced feature of the Johnson and Constable (1995) time-averaged field model for the last 5 Myr is an inclination anomaly during reversed states of the geomagnetic field in the Equatorial and South Atlantic Ocean, which is accompanied by a declination anomaly. The anomaly in the model of Johnson and Constable (1995) heavily relies on five reversely magnetized sampling sites located on Fernando de Noronha (Richardson and Watkins, 1967). A distinct inclination anomaly over Fernando de Noronha is also found in the present day field geomagnetic field (e.g. IGRF 2000).

It is the purpose of this work to extend the existing paleomagnetic data from Fernando de Noronha (Richardson and Watkins, 1967; Mazoires et al., 1971; Schult et al., 1986) in terms of paleodirections and absolute paleointensities. Samples from 44 volcanic units were obtained and analyzed for rock magnetic and paleomagnetic characteristics. Directional results are compared to previously published data and we scrutinize normal versus reverse field asymmetry, as well as persistent long-term anomalies of the geomagnetic field for Fernando de Noronha. Absolute paleointensity determinations were conducted using a modified Thellier-type technique, which allow for the recognition of possible biases by multidomain (MD) remanence. This methodology provides a quality check regarding correction methods for magnetomineralogical changes which occur during laboratory treatment. Finally, our paleointensity results are compared to published intensity values of the same time range.

2. Geology and sampling strategy

Fernando de Noronha island is about 365 km northeast of Natal, Rio Grande do Norte, Brazil on 3.85° south and 32.4° west. The island constitutes the eastern end of a chain of volcanic sea-mounts which are thought to be of hotspot origin (e.g. Rivalenti et al., 2000). Basically, two main volcanic formations are defined on the island (Fig. 1). The main rocks of the Quixaba formation (3.3–1.8 Myr) are ankaratritic lavas of similar composition indicating a common mantle source (Ulbrich et al., 1994). The Remedios formation with an average age of 9.5 ± 1.1 Myr consists of phonolites, nephelinites, alkali trachytes, and alkali basalts. Age determinations were conducted by Cordani (1970) using the K/Ar technique. Samples from 44 sites of both formations were obtained predominantly from coastal outcrops where successions of lava flows are exposed (Fig. 1). From the Quixaba formation 27 volcanic units were sampled, 25 ankaratritic lava flows, one dyke (site FN-Q16) and a nepheline basanite from Illa Cuscus (site FN-Q1). Seventeen volcanic units were sampled in the Remedios formation. Sites FN-R6, FN-R7, FN-R9, FN-R5 (8.8 Ma), FN-R3 and FN-R8 are phonolitic units. The alkali trachyte of site FN-R13 has an age of 10.8 Ma. Four lava flows (FN-R14 to Fn-R17) were sampled in approximately 12.3 Myr old alkali basalts. In addition, samples from one glenmuirite site (FN-R4), two dykes (FN-R1, FN-R2) and essexites (FN-R10 to FN-R12, 9.4 Myr) were taken in the Remedios formation. From each site at least five separately oriented drill cores were taken, oriented with a magnetic and a sun compass. Whenever possible, the individual samples of one unit were taken with horizontal spacings of several meters and different vertical positions. There are no indications for major tectonic movements like rotations, faulting and unconformities in the sampled volcanic units on Fernando de Noronha, wherefore no tectonic corrections are applied.

3. Methodology and laboratory analysis

Measurements and demagnetization of the natural remanent magnetization (NRM) were performed in the magnetically shielded room at the paleomagnetic laboratory of the University of Munich in

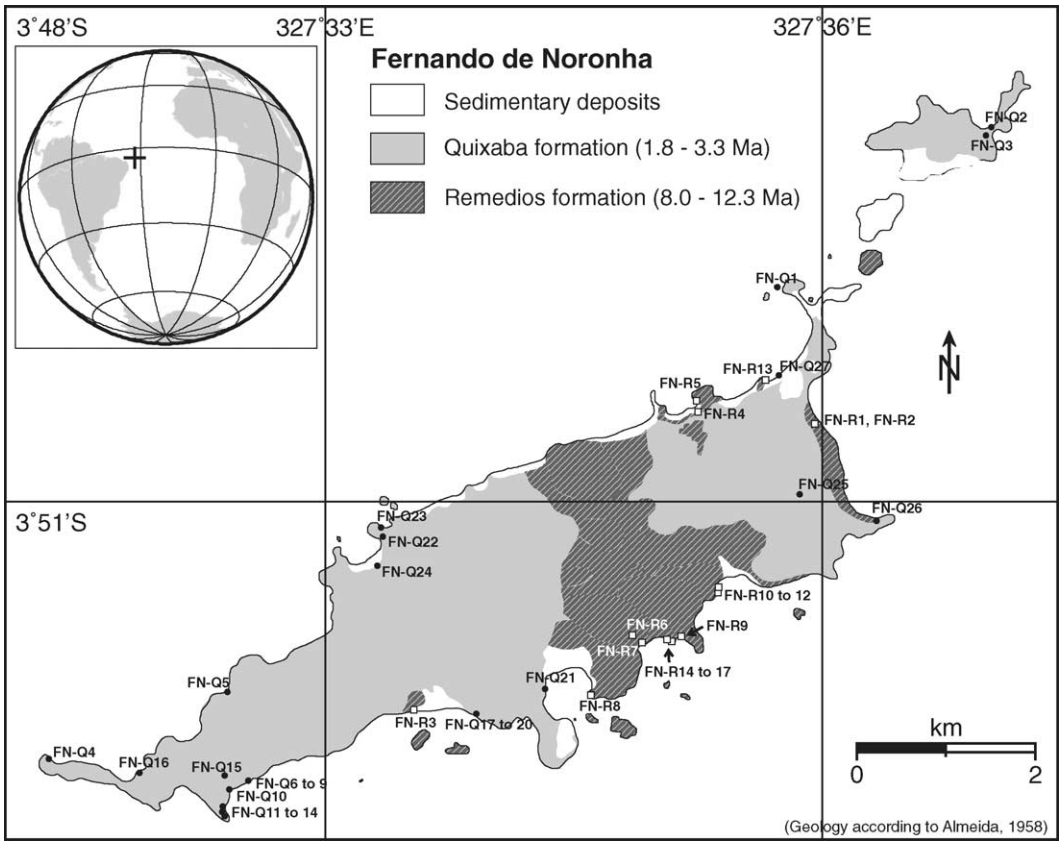


Fig. 1. Geological sketch map of Fernando de Noronha after Almeida (1958). Points indicate sampling sites in the Quixaba formation, white rectangles are sampling sites in the Remedios formation.

Niederlippach. Remanence measurements were carried out using a Molspin spinner magnetometer for strongly magnetized samples and a 2G cryogenic magnetometer for weakly magnetized samples. On average, one-half of the 249 samples were thermally demagnetized up to 680 °C and the other half with stepwise alternating field (AF) demagnetization up to 150 mT. The demagnetization data were analyzed using principle component analysis. For the majority of the lava flows, thermal and AF demagnetization were equally effective for determining the characteristic remanent magnetization (ChRM) of the samples. The low-field magnetic susceptibility was measured at room temperature for every sample and after each thermal demagnetization step using a KLY-1 kappa bridge. Continuous demagnetizations were performed with a high temperature spinner magnetometer (HOT-SPIN, described by Matzka et al. (2003)) which measures simultaneously two orthogonal components of the NRM at temperatures of up to 600 °C. At least one sample from every flow was used for a set of rock magnetic measurements. We measured IRM-acquisition, backfield curves, hysteresis loops at room temperature and thermomagnetic curves ($B_{max} = 900$ mT, $T_{max} = 700$ °C) for each of these samples using a Variable Field Translation Balance (VFTB). Additionally, 19 samples from reverse and normal states of both formations were examined using ore microscopy.

4. Rock magnetics and ore microscopy

Analysis of the hysteresis measurements from samples of the Quixaba formation according to Day

et al. (1977) indicates a grain size distribution of the magnetic particles in the pseudo-single-domain range (M_{rs}/M_s = 0.07–0.39, B_{cr}/B_c = 1.5–3.1). Strong field thermomagnetic curves in fields of 550 mT were obtained from most sites in order to determine the magnetic phases in the samples. The magnetic intensities of 3 units from the Remedios formation were below the sensitivity of the VFTB. Two thermomagnetic curves are shown in Fig. 1. Based upon Curie temperature T_C, two different types of thermomagnetic behavior were observed. Samples from 15 units of the Quixaba formation and 6 units of the Remedios formation yielded $M_s(T)$-curves characterized by a single ferromagnetic phase with Curie temperatures between 500 and 560 °C (Fig. 2a), indicative of a low-Ti titanomagnetite. Ore microscopy shows that this magnetic phase evolved from the primary Ti-magnetite by the migration of titanium to form ilmenite lamellae during high temperature oxidation during cooling of the lava flow (Fig. 3a). In addition to this phase, hematite is present in 4 lava flows, marked by Curie temperature of ~670 °C. During all thermal demagnetization experiments, the hematite component revealed the same direction as the low-Ti titanomagnetite component suggesting that hematite formed during or shortly after primary cooling of the lavas.

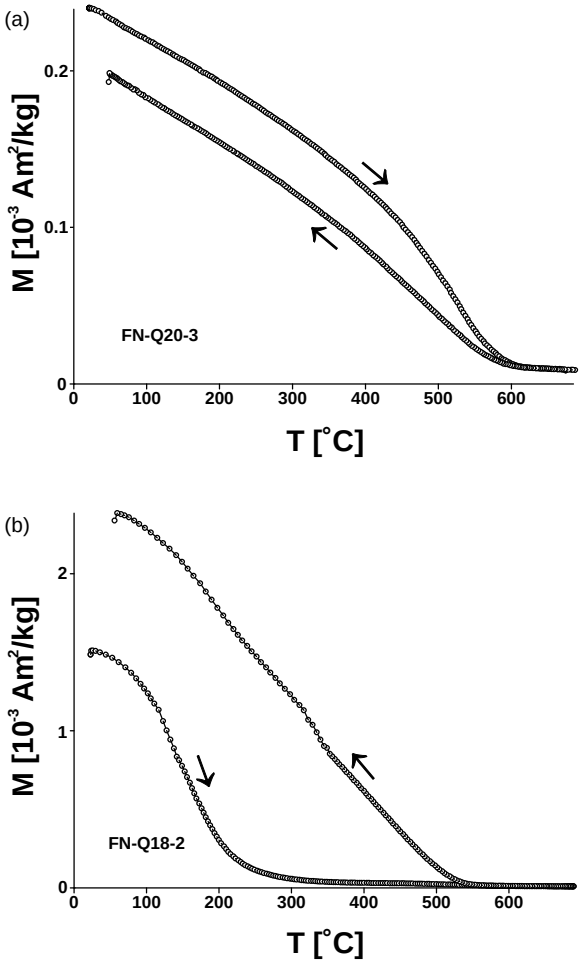


Fig. 2. Representative examples of thermomagnetic curves from samples of the Quixaba formation.

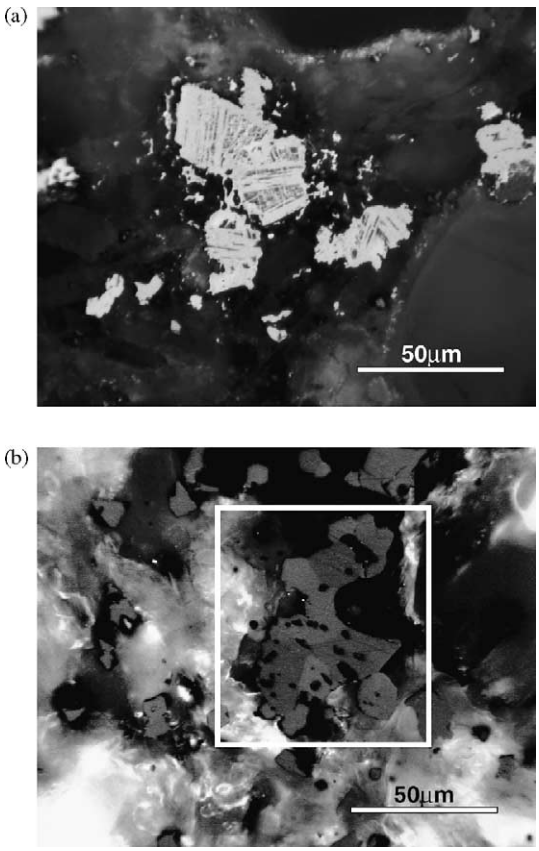


Fig. 3. Two ore microscopy pictures. (a) Corresponds to sample FN-Q19-2 and is covered by a magnetic colloid in order to emphasize the ilmenite lamellae formed during high temperature oxidation. (b) Shows an inhomogeneous grain of sample FN-Q23-2 in polarized light. This sample also exhibits partial self-reversal.

In addition to low-Ti titanomagnetites, a second ferrimagnetic phase with a T_C-range from 100 to 400 °C (Fig. 2b) is observed in 12 units of the Quixaba formation and 8 units of the Remedios formation. Ore microscopy revealed titanomagnetites of different grades of alteration within these samples. In particular, the alkali basaltic samples from the Remedios formation are highly altered. Some samples show predominantly inhomogeneous grains (Fig. 3b). Interaction of magnetic phases with different Curie temperatures in one grain may result in partial self-reversal during laboratory treatment. Several samples were subjected to continuous demagnetization, where the blocking and unblocking of remanences can be observed directly and continuously, allowing magnetostatic interaction to be monitored. The subject of partial self-reversal exceeds the purpose of this paper and will be discussed in detail in Wehland et al. (submitted). Partial self-reversal of the natural remanent magnetization was found in four samples of the Quixaba formation and one sample of the Remedios formation, characterized by two T_C. Due to the directional influence of the interaction, site FN-Q4 was rejected from further analysis.

5. Paleomagnetic results

5.1. Paleodirections

Thermal and AF demagnetization were equally effective in determining the characteristic remanent

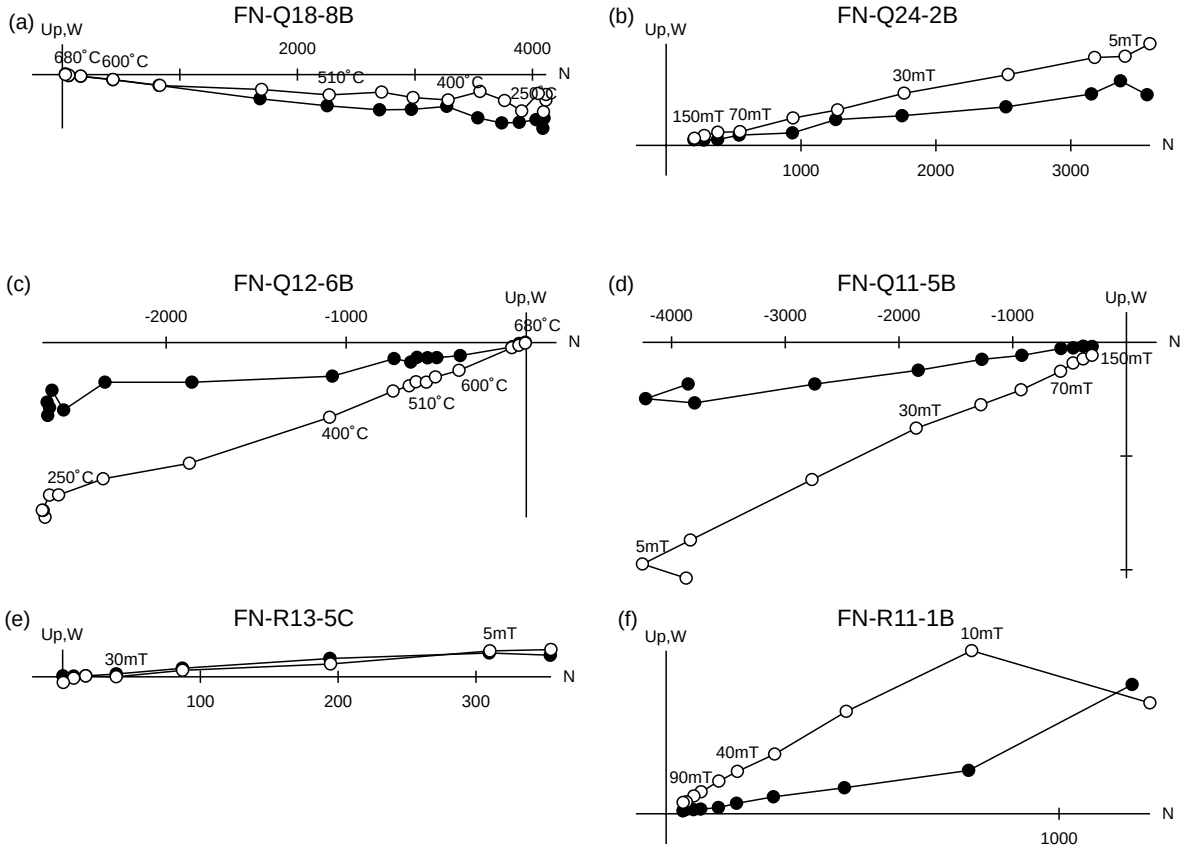


Fig. 4. Representative orthogonal projections of thermal and AF demagnetizations form normally and reversely magnetized lava flows from the Quixaba formation (a–d) and normally magnetized lavas from the Remedios formation (e, f). Open symbols correspond to projections on the vertical plane, solid symbols to the projection on the horizontal plane.

Table 1
Paleodirectional results from Fernando de Noronha[a]

Site	$Site_{lat}$ (°)	$Site_{long}$ (°)	$n/N(N_{gc})$	P	Declination (°)	Inclination (°)	k	α_{95} (°)	DG	$Declination_{DG}$ (°)	$Inclination_{DG}$ (°)	k_{DG}	α_{95} (DG) (°)
Quixaba formation													
FN-Q1$^{(59S)}$	−3.8283	327.5955	6/5(1)	N	9.1$^{(2.7)}$	−15.8$^{(-18.3)}$	287	4.7$^{(2.6)}$	FN-QDG1	9.1	−15.8	286.6	4.7
FN-Q2	−3.8120	327.6170	5/5	N	4.0	−21.9	96	7.8	FN-QDG2	4.0	−21.9	96.2	7.8
FN-Q3	−3.8130	327.6165	6/5(2)	N	8.4	−31.9	77	9.4	FN-QDG3	8.4	−31.9	77.3	9.4
FN-Q4	−3.8762	327.5222	5/0										
FN-Q5	−3.8693	327.5403	5/5	R	169.0	23.6	170	5.9	FN-QDG5	169.0	23.6	169.7	5.9
FN-Q6	−3.8783	327.5422	6/6	R	172.4	15.5	169	5.2	FN-QDG6	172.4	15.5	168.8	5.2
FN-Q7	−3.8783	327.5422	7/6	R	175.6	20.8	425	3.3					
FN-Q8	−3.8783	327.5422	6/6	R	171.4	21.2	211	4.6					
FN-Q9	−3.8783	327.5422	6/6	R	170.4	20.1	276	4.0	FN-QDG7	172.9	21.9	291.7	1.6
FN-Q10$^{(1R)}$	−3.8791	327.5403	6/6	R	173.5$^{(170.5)}$	23.8$^{(24.7)}$	341	3.6$^{(4.5)}$					
FN-Q11$^{(3R)}$	−3.8812	327.5398	5/5	R	173.8$^{(172.6)}$	24.1$^{(26.5)}$	1515	2.0$^{(4.3)}$					
FN-Q12	−3.8812	327.5398	6/6	R	172.5	17.8	421	3.3					
FN-Q13$^{(5R)}$	−3.8818	327.5398	5/5	R	172.6$^{(169.9)}$	19.0$^{(23.8)}$	1455	2.0$^{(3.1)}$					
FN-Q14	−3.8815	327.5398	5/5	R	168.8	15.5	965	2.5	FN-QDG9	168.8	15.5	964.8	2.5
FN-Q15	−3.8778	327.5400	5/5	R	189.5	25.2	380	3.9	FN-QDG10	189.5	25.2	379.6	3.9
FN-Q16	−3.8775	327.5315	5/5	R	171.6	13.0	730	2.8	FN-QDG11	171.6	13.0	729.6	2.8
FN-Q17	−3.8715	327.5653	6/6	N	7.8	10.7	148	5.5					
FN-Q18	−3.8715	327.5653	8/6	N	7.3	5.0	168	5.2	FN-QDG12	7.4	6.5	145.8	2.9
FN-Q19	−3.8715	327.5653	6/6	N	7.2	3.8	177	5.1					
FN-Q20	−3.8715	327.5653	6/6	N	14.1	5.4	705	2.5	FN-QDG13	14.1	5.4	704.5	2.5
FN-Q21$^{(9R)}$	−3.8688	327.5722	5/5	N	355.3$^{(350.6)}$	−5.4$^{(-4.8)}$	440	3.7$^{(7.5)}$	FN-QDG14	355.3	−5.4	439.5	3.7
FN-Q22	−3.8535	327.5557	6/6	N	355.0	−13.2	130	5.9	FN-QDG15	355.0	−13.2	130.1	5.9
FN-Q23	−3.8525	327.5557	6/6	N	355.2	−13.5	214	4.6	FN-QDG16	355.2	−13.5	214.2	4.6
FN-Q24	−3.8565	327.5554	5/5	N	352.1	−13.6	435	3.7	FN-QDG17	352.1	−13.6	434.9	3.7
FN-Q25	−3.8492	327.5978	5/5	N	15.4	−3.3	189	5.6	FN-QDG18	15.4	−3.3	189.3	5.6
FN-Q26	−3.8520	327.6053	6/6	N	7.2	−17.8	265	4.1	FN-QDG19	7.2	−17.8	265.1	4.1
FN-Q27$^{(58S)}$	−3.8372	327.5957	7/6	N	353.8$^{(350.1)}$	−7.0$^{(-3.8)}$	120	6.1$^{(2.3)}$	FN-QDG20	353.8	−7.0	120.1	6.1
Remedios formation													
FN-R1	−3.8423	327.5999	5/5	N	20.5	−1.9	988	2.4	FN-RDG1	20.5	−1.9	988.3	2.4
FN-R2	−3.8423	327.5999	5/4	N	4.3	−19.3	478	4.2	FN-RDG2	4.3	−19.3	478.2	4.2
FN-R3$^{(6R)}$	−3.8712	327.5588	5/5	N	13.3$^{(355.7)}$	−29.5$^{(-33.3)}$	376	4.0$^{(40.7)}$	FN-RDG3	13.3	−29.5	376.1	4.0
FN-R4	−3.8408	327.5875	5/5(1)	R	157.5	30.0	83	8.7	FN-RDG4	157.5	30.0	82.5	8.7
FN-R5	−3.8398	327.5875	5/0										
FN-R6	−3.8635	327.5810	5/5(4)	I	57.0	−31.0	119	8.8	FN-RDG5	57.0	−31.0	119.2	8.8
FN-R7	−3.8640	327.5820	6										
FN-R8$^{(11R)}$	−3.8698	327.5768	6/6(6)	R	187.3$^{(53.0)}$	24.8$^{(77.2)}$	180	6.5$^{(18.4)}$	FN-RDG6	187.3	24.8	180.1	6.5

FN-R9	−3.8640	327.5858	6/5	N	359.2	−7.2	539	3.3	FN-RDG7	359.2	−7.2	539.1	3.3
FN-R10$^{(17R)}$	−3.8587	327.5897	6/6(1)	N	355.7$^{(355.9)}$	−35.3$^{(-33.9)}$	197	4.9$^{(3.7)}$					
FN-R11	−3.8593	327.5897	6/6	N	352.5	−30.8	182	5.0	FN-RDG8	352.8	−33.8	147.7	2.9
FN-R12	−3.8587	327.5897	6/6	N	350.2	−35.2	116	6.3					
FN-R13$^{(22R)}$	−3.8377	327.5945	5/5	N	357.5$^{(353.6)}$	3.8$^{(-1.3)}$	268	4.7$^{(11.9)}$	FN-RDG9	357.5	3.8	268.1	4.7
FN-R14	−3.8640	327.5842	5/0										
FN-R15	−3.8640	327.5845	6/0										
FN-R16	−3.8640	327.5849	6/6(1)	R	173.0	−23.7	95	7.0	FN-RDG10	173.0	−23.7	95.3	7.0
FN-R17	−3.8640	327.5852	6/6(4)	R	172.7	−15.1	129	6.6	FN-RDG11	172.7	−15.1	129.3	6.6

[a]Shown are site mean inclinations and declinations, as well as the precision parameter k and α_{95} for individual sites and directional groups DG. n/N denotes the number of samples treated (n) versus the number of samples used (N) for calculating the site mean direction. The number of samples analyzed with great circles (N_{gc}) is given in brackets. Also shown are the site coordinates and the polarity P (R: reversed, N: normal, and I: intermediate if the difference between Earth's rotation axis and VGP latitude is >30°). Superscripted annotations to the site name indicate re-sampled sites from previous studies. 59S, for example, indicates that samples were taken from the same rock unit, which is published as site 59 in Schult et al. (1986), 1R corresponds to site 1 of Richardson and Watkins (1967). Inclination, declination and α_{95} of the previous studies are shown in parentheses in the corresponding column.

magnetization (ChRM). Typical demagnetization diagrams from the Quixaba and Remedios formation are shown in Fig. 4. In both formations normally and reversely magnetized units were sampled. Most of the samples from the Quixaba formation are characterized by a single stable component without viscous overprints (Fig. 4a–d), typified by univectorial behavior during demagnetization. Viscous overprints were observed, especially for samples from the Remedios formation, but these were usually removed by 250 °C or 30 mT.

For the majority of the samples from the Quixaba formation the characteristic mean direction could be identified using principle component analysis. For five samples stable end point analysis was not suitable and the great circle technique (GCA) according to McFadden and McElhinny (1988) was applied. A few samples with unstable directions were excluded from the analysis. The mean directions for each flow were calculated using Fisher (1953) statistics (Table 1). Overall the within-site scatter is small as in more than half of the sites we find $\alpha_{95} < 5^\circ$, $\alpha_{95} < 10^\circ$ in the remaining, coupled with k values above 100 for the majority of the Quixaba sites, all underlining the high quality of the directional data. Samples from the Remedios formation (Fig. 4e and f) show more complex demagnetization behaviors. The alkali basalts (units FN-R14 and FN-R15), in particular, and the weakly magnetized phonolites of unit FN-R5 and FN-R7 do not show a coherent demagnetization data and could not be used for further analysis (see Table 1). It proved necessary to employ GCA for analysis of the demagnetization data of unit FN-R8. NRM intensities for the majority of the sites of the Remedios formation are much lower than those observed for the Quixaba formation, except site FN-R1 which also shows partial self-reversal. FN-R1 corresponds to a dark colored dyke, rich in olivine crystals, cutting an older dyke FN-R2. Based on the composition and the strong intensity signal, the dyke FN-R1 is thought to be part of the Quixaba formation.

Several sites on the western side of the island show similar directional records (Capim Açu section, FN-Q6 to FN-Q14). The absence of markers indicating periods of volcanic inactivity, such as paleosoils or erosional unconformities, suggest that most lava flows of this section were extruded within a short period of time. Similar observations and absence of non-ambiguous flow boundaries apply for lava flows of the Leão Beach section (FN-Q17 to FN-Q20).

In order to quantify whether the sampled lavas recorded independent states of the geomagnetic field, comparison of directions using the *F*-distribution test (e.g. Butler, 1992), were conducted. The probability to which successive flows are independent was calculated and for values less than 95% between flows, the lavas are assumed to have recorded the same geomagnetic field. Calculation started from the lowest and, therefore, oldest flow. *F*-distribution statistics for the Capim Açu section support the presence of five different directional records of the past geomagnetic field, whereas only two were isolated for the Leão Beach section. Non-independent sites were combined to directional groups (Table 1). This purely directional approach to serial correlation is supported by paleointensity determinations of samples from the Leão Beach section, where flow FN-Q18 and FN-Q19 show almost identical paleointensity values.

Ten volcanic units, whose paleomagnetic results are presented in this study, were previously sampled and analyzed by Richardson and Watkins (1967) and Schult et al. (1986). These include six lavas from the Quixaba formation and four from the Remedios formation (Table 1). For the Quixaba formation our paleomagnetic data is in good agreement with the results from both previous studies. Significant differences are only observed for the very weakly magnetized phonolites of site FN-R3 and FN-R8. These two sites are marked by high to very high α_{95}-values of 18.4° and 40.7° in the earlier investigations.

6. Paleointensity determination

6.1. Selection criteria

Only samples yielding consistent within-site ChRM directions were used for paleointensity determinations. Suitable samples for Thellier-type paleointensity determinations were selected based upon the results of rock magnetic examinations and ore microscopy observations. Samples showing a single high Curie temperature, indicative of high temperature oxidation during flow emplacement, were used as they are considered to be insensitive to further oxidation under laboratory conditions. As thermomagnetic

curves were not measured from every sample of one site and since standard and continuous demagnetization experiments showed that within-site differences are common, additional pre-selection methods were applied. Low-field susceptibility measurements after each thermal demagnetization step allow the estimation of the thermal stability of the samples. Samples with only slight variations are termed to be suitable. Twenty-nine specimens were subjected to AF demagnetization of an anhysteretic remanent magnetization ARM (150 mT AC, 50 μT DC) before and after heating to 200, 400 and 550 °C, respectively. These measurements, which are similar to the alteration analysis of Shaw (1974), provide a accurate quantification of possible alteration mechanisms. Two examples of these experiments are shown in Fig. 5 wherein AF demagnetization of an ARM at room temperature is plotted versus the AF demagnetization of an ARM after several heating steps. If the ARM demagnetizations following subsequent heating steps are linearly related with a slope of one, then no alteration occurred (Fig. 5a). Irreversible changes of the magnetic information due to alteration at moderate temperatures (Kosterov and Prévot, 1998) as well as high temperature alteration can be detected. Alteration at low temperatures (e.g. stress release in titanomagnetites), which could lead to spurious paleointensity estimates, are normally not detected using standard checks during the Thellier experiment, as only a small percentage of the NRM is lost below 300 °C. Fig. 5b shows a gradual alteration of a sample during laboratory treatment at temperatures above 200 °C. The low coercivity fraction increases during heating, whereas the high coercivity fraction decreases. Such samples are not suitable for paleointensity determinations.

Only samples from 12 sites of the Quixaba formation passed the selection criteria for paleointensity determination.

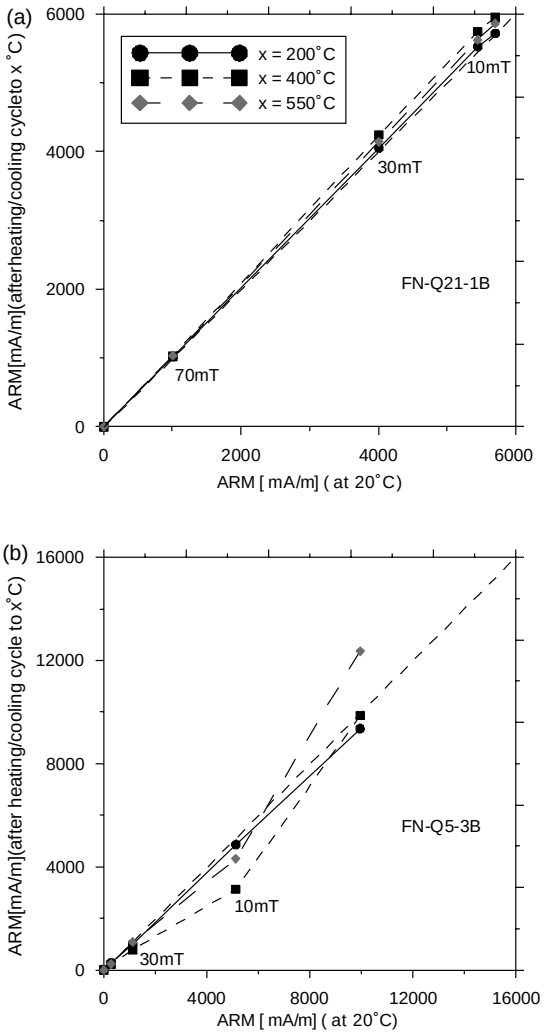


Fig. 5. AF demagnetization of an ARM at room temperature versus the AF demagnetization after heating to 200, 400, and 550 °C.

6.2. Methodology of paleointensity determination

All paleointensity determinations were performed in an argon atmosphere on minicores with an diameter of 5 mm in an applied laboratory field of 30 μT. The minicores were taken unoriented from the lower part of individual drill cores and the primary character of the direction was ascertained by demagnetization of sister specimens from the same drill core. Due to the smallness of the specimens the orientation along the *x*- and *y*-axis within the cryogenic magnetometer is afflicted with slightly higher errors, leading to a more scattered declination information than for inch specimens. A modified Thellier technique was used on all samples. pTRM checks, after Coe (1967), were performed after every second heating step in an effort to monitor alteration during laboratory treatment. Repeated demagnetization steps following pTRM acquisition to the same temperature were conducted and facilitate the determination of multidomain (MD) tails (Riisager and

Riisager, 2001), thermal stabilization processes, and/or alteration affecting unblocking temperatures above the actual heating step. MD tails of pTRM as well as the formation of new magnetic material during the pTRM step will lead to an increase of the magnetization in the direction of the applied field, which is parallel to the *z*-component, in sample coordinates, in our experiments. Thermal stabilization processes cause a decrease of the magnetization during repeated heating steps to the same temperature and do not affect the direction of the repeated demagnetization step (McClelland and Briden, 1996). In order to detect differences between blocking (T_b) and unblocking (T_{ub}) temperatures, indicative of MD remanence, the additivity of pTRMs was checked during the experiment (Krása et al., 2003). For this purpose a pTRM(T_i,T_0) is demagnetized thermally up to T_k ($k < i$) and the remaining remanence M_{rem} is then measured. In the case of pure single-domain (SD) remanence:

$$M_{rem} \triangleq M_{pTRM}(T_i, T_0) - M_{pTRM}(T_k, T_0). \quad (1)$$

If the remanence is carried by MD particles M_{rem} will be less than that difference. This check is sensitive to MD remanences with $T_{ub} < T_b$, which cause curvature in the NRM/pTRM plots (Dunlop and Özdemir, 2000; Fabian, 2001). Similar to the pTRM check, the additivity check (AC) is sensitive to alteration during the laboratory treatment. In cases of single-domain (SD) particles ($T_{ub} = T_b$) and no ongoing alteration during repeated heating steps, pTRM checks and ACs conducted over the same temperature range give the same results.

6.3. Paleointensity determinations

Specimens from 12 of the 44 sampled volcanic units were selected and subjected to paleointensity determinations. Altogether, 42 successful determinations were obtained from 49 specimens indicative of an accurate pre-selection procedure. Representative examples of Thellier-type paleointensity determinations are shown in Fig. 6. Analysis of paleointensity determinations was performed using standard NRM/pTRM plots. For identification of linear segments within NRM/pTRM diagrams only successive points were used.

6.4. Corrections for magnetomineralogical changes

If alteration checks are continuously performed over the entire experimental temperature range and assuming that alteration happens predominantly below the temperature of the respective check, a method proposed by Valet et al. (1996) can be used to correct for magnetomineralogical changes during laboratory treatment. A fundamental requisite for this correction method is the absence of MD remanence. As shown above, the additivity checks are similar to the standard pTRM checks, with an added sensitivity to $T_{ub} < T_b$. Therefore, one can use the method of Valet et al. (1996) to correct the paleointensity determination using the pTRM checks. For this technique, the cumulative sum of the individual differences between checks and associated pTRM values up to heating step $i - 1$ is subtracted from the measured pTRM value at step i. After applying the alteration correction, the additivity checks should fall on the corrected pTRM values for cases of SD remanence. If this is not the case, a significant bias from MD remanence or continuous alteration during repeated heating steps is present. Two examples of alteration corrected results are shown in Fig. 7. Fig. 7a shows a slightly downward curved NRM/pTRM plot. Neither the repeated demagnetization steps nor the ACs show a significant influence of MD remanence. After check correction, the fraction of the NRM covered by the linear segment is well above 50%. A similar paleointensity result is obtained for the non-corrected analysis. In Fig. 7b an example is shown for which it is not possible to analyze the non-corrected determination. After check correction, the ACs coincide with the pTRM values and a linear segment covering most of the blocking temperature spectrum is obtained. Twenty specimens of the collection were analyzed using check correction. For seven of these, analysis without check correction would have failed.

6.5. Analysis criteria and results

The following criteria were used to characterize the quality of the absolute paleointensity values from the NRM/pTRM plots:

(1) The temperature range used for the calculation of a linear fit must be related to the characteristic

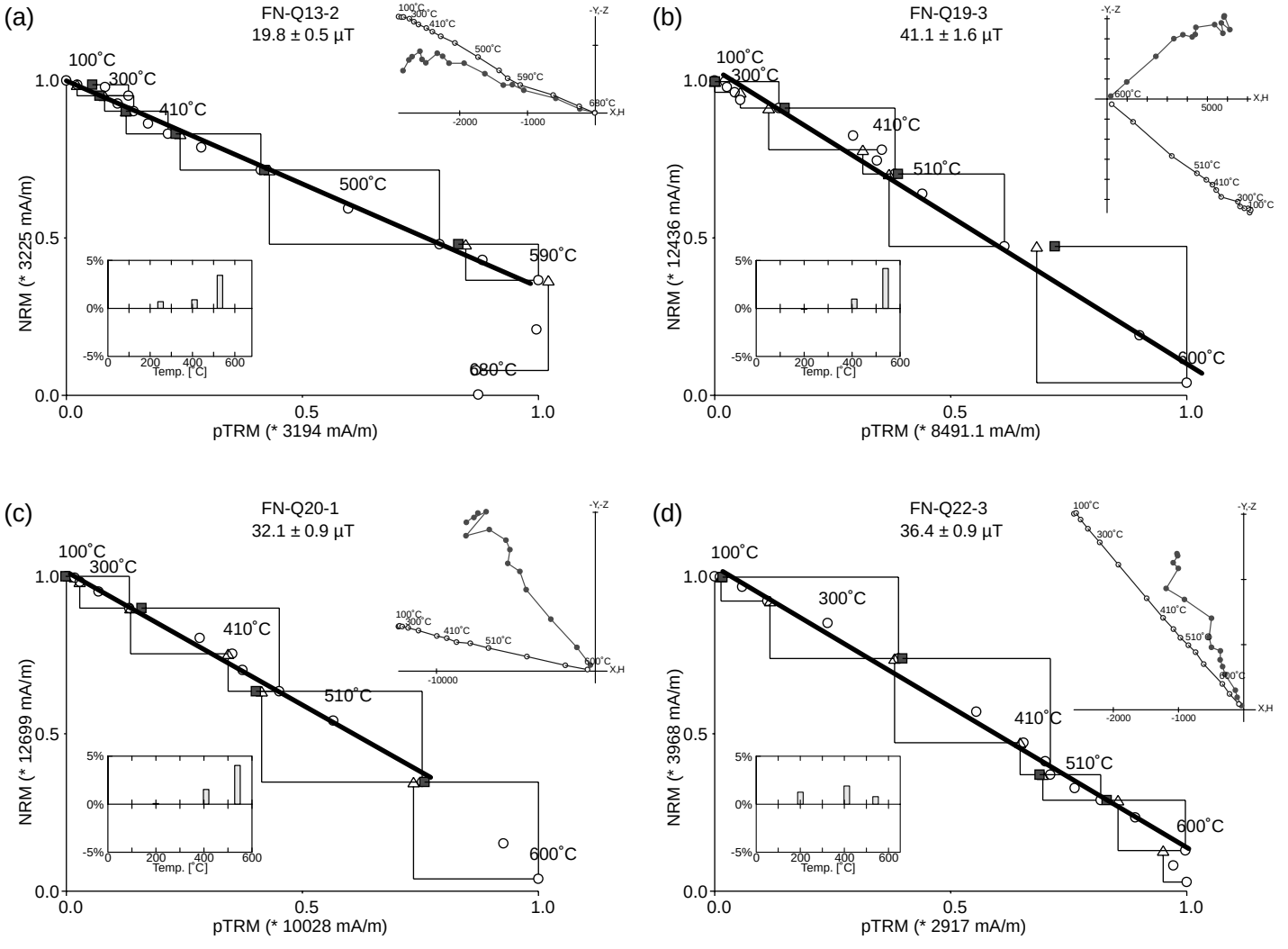


Fig. 6. NRM/pTRM diagrams for four samples of the Quixaba formation. The straight line segment is well defined and covers more than 50% of the NRM in 70% of the collection. Triangles indicate pTRM checks and squares denote additivity checks. The inset on the lower left of each plot shows the relative intensity difference of repeated demagnetization steps normalized to the true NRM. The directional demagnetization behavior is shown in the orthogonal projection in the upper right of each plot.

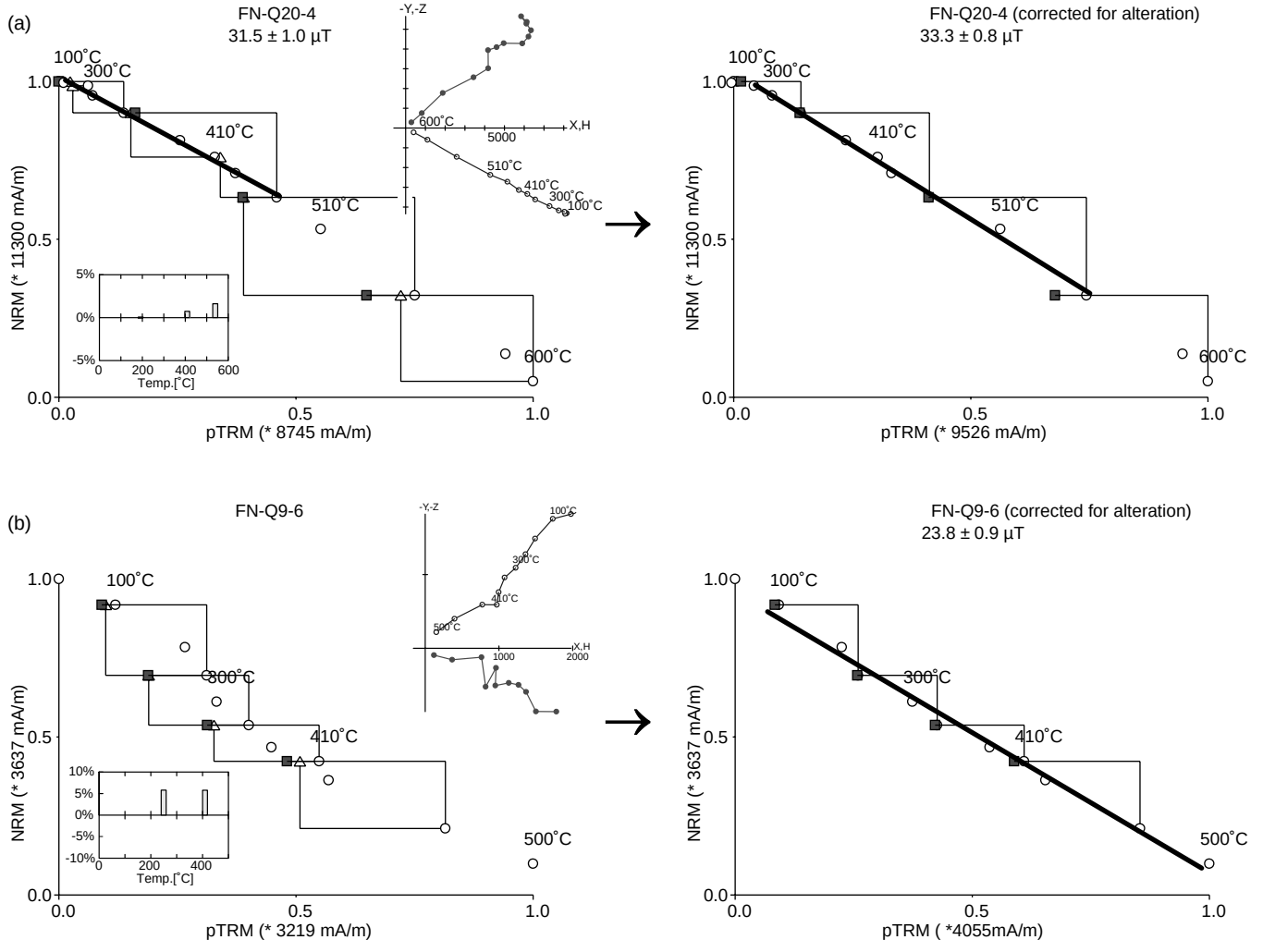


Fig. 7. NRM/pTRM diagram for two samples of the Quixaba formation. The plots on the left side show the uncorrected paleointensity determination. When applying the correction method of Valet et al. (1996) the plots on the right side are obtained. The ACs now fit the corrected pTRM values indicating that MD remanence does not bias the determination and the curvature of the original data is the result of alteration.

remanent magnetization of the sample. Therefore the maximum angular deviation (MAD) of the direction of the used segment has to be less than 6°.

(2) The segments used for the linear fits consist of at least five successive points.

(3) The segments cover a fraction (f) of at least 30% of the NRM.

(4) Only limited alteration has occurred during the experiment. In order to quantify the effect of alteration during laboratory treatment, two parameters were calculated:

   (a) The first parameter, δ_{CK}, gives a relative measure of the individual alteration effect at a specific temperature. The difference between pTRM check and related pTRM acquisition is normalized to the true TRM, which is the intersection of the best fit line and the pTRM-axis. This method does not overemphasize low temperature steps, where the pTRM acquired is rather small. Additionally, it does not tolerate large check errors if the selected segment is long, as it is the case when using the length of the selected segment as normalizer (Selkin and Tauxe, 2000). δ_{CK} must not exceed 5% before and within the linear segment.

   (b) The second parameter, δ_{pal}, is related to the cumulative difference of the individual checks from room temperature up to the maximum temperature used for the best fit line. This parameter estimates the overall alteration, as even small individual differences between checks and pTRM values can sum to significant alteration errors. To quantify the effect of the cumulative alteration difference, the correction method of Valet et al. (1996), which uses the cumulative sum of alteration differences, is applied to the selected segment and the check corrected paleointensity value is compared to the non-corrected value. These values should not differ by more than 5% with respect to the non-corrected value.

   In case of alteration corrected analysis, δ_{CK} and δ_{pal} are not used as criteria for the determination.

(5) The scatter of the data points about the linear fit should be small. The standard deviation of the paleointensity determination must be $\leq$10%.

(6) To exclude any significant bias because of MD tails, the intensity change between the demagnetization step preceding (NRM(T_i)) and the step succeeding (NRM$_{re}$(T_i)) the pTRM acquisition must not exceed 5%.

(7) Additivity checks must not differ more than 5% from the associated pTRM value normalized to the true TRM. Hereafter, the AC difference is named δ_{AC}. In cases of check corrected analysis, δ_{AC} must not differ more than 5% from the associated alteration corrected pTRM value.

Samples which meet all these criteria were termed to be class A determinations. The limiting values for class B determinations are that $\delta_{CK} < 7\%$, the standard deviation $< 15\%$ and the MAD $< 10°$. Determinations were accepted only when one of these values exceed the class A criteria. Additionally, the intensity change between NRM(T_i) and NRM$_{re}$(T_i) has to be $<$10% and $\delta_{AC} < 10\%$. Specimens exceeding these values and with $f < 0.3$ and $N < 5$ were not accepted. If the alteration correction is used for paleointensity determination, a star is added to the class.

From each volcanic unit, several specimens from different drill cores were analyzed for paleointensities. Weighted mean averages, as well as weighted standard deviations, were calculated using the weighting (w) factor (Prévot et al., 1985). The results and statistical parameters are summarized in Table 2.

7. Discussion

7.1. Directional aspects

The distribution of normal and reverse directions from the Quixaba formation show two distinct areas. Reversely magnetized units were found only on the western side of Fernando de Noronha. K/Ar age determinations (Cordani, 1970)yield ages of about 2.5 Myr for these volcanic units, emphasizing that this part of the island formed during the early Matuyama polarity chron. All other Quixaba samples from the main island show normal magnetizations with associated ages of about 3.0 Myr and were extruded during the Gauss polarity chron. Two sites (FN-Q2 and FN-Q3) on Ilha Rata, northeast of Fernando de Noronha, also show normal directions. According to Cordani (1970) the

Table 2
Paleointensity results from Fernando de Noronha[a]

Site	*n*/*N*	Specimen	T_{min} (°C)	T_{max} (°C)	Np	*f*	*g*	*q*	*w*	C	*H* ± S.D. (μT)	F_w ± S.D.$_w$ (μT)
FN-Q3	2/5	FN-Q3-3	380	540	6	0.50	0.64	2.9	1.5	B	7.5 ± 0.8	9.6 ± 0.6
		FN-Q3-4	440	590	6	0.70	0.73	36.1	18.1	A*	9.7 ± 0.1	
FN-Q4	0/1											
FN-Q9	5/5	FN-Q9-1	20	510	11	0.96	0.83	12.7	4.2	B*	27.1 ± 1.7	24.4 ± 1.1
		FN-Q9-2	20	340	6	0.54	0.76	5.3	2.6	A*	26 ± 2.0	
		FN-Q9-4	100	440	8	0.66	0.85	12.0	4.9	B*	25.9 ± 1.2	
		FN-Q9-5	20	530	12	0.95	0.89	17.4	5.5	A	21.1 ± 1.0	
		FN-Q9-6	100	500	10	0.85	0.87	19.0	6.7	B*	23.8 ± 0.9	
FN-Q13	5/5	FN-Q13-1	200	410	6	0.32	0.75	3.0	1.5	B	24.1 ± 1.9	20.2 ± 1.3
		FN-Q13-2	20	530	12	0.53	0.85	14.9	4.7	A	20.5 ± 0.6	
		FN-Q13-3	380	590	8	0.35	0.80	6.2	2.5	A*	24.5 ± 1.1	
		FN-Q13-4	200	510	9	0.47	0.84	12.3	4.6	A	17.1 ± 0.6	
		FN-Q13-5	200	570	11	0.49	0.84	13.3	4.4	B	19.4 ± 0.6	
FN-Q15	4/4	FN-Q15-1	20	300	5	0.52	0.73	6.6	3.8	B	55.4 ± 3.2	42.5 ± 7.4
		FN-Q15-2	100	440	8	0.55	0.82	5.9	2.4	B*	41.2 ± 3.2	
		FN-Q15-3	200	440	7	0.30	0.81	7.6	3.4	A*	25.1 ± 0.8	
		FN-Q15-4	20	480	10	0.34	0.84	3.7	1.3	B*	52.9 ± 4.1	
FN-Q16	4/5	FN-Q16-1	20	480	10	0.60	0.85	7.7	2.7	B*	33.5 ± 2.2	28.2 ± 2.3
		FN-Q16-2	20	480	10	0.60	0.86	8.3	2.9	B*	29.9 ± 1.9	
		FN-Q16-3	200	480	8	0.46	0.82	5.7	2.3	B*	26.2 ± 1.7	
		FN-Q16-5	100	470	9	0.47	0.86	7.7	2.9	A*	23.2 ± 1.2	
FN-Q18	1/1	FN-Q18-8	100	440	8	0.33	0.85	7.8	3.2	B	38.8 ± 1.4	
FN-Q19	4/4	FN-Q19-2	20	600	14	0.93	0.88	26.8	7.7	B	38.8 ± 1.2	41.1 ± 1.9
		FN-Q19-3	20	600	14	0.92	0.84	19.9	5.7	B	41.1 ± 1.6	
		FN-Q19-4	440	600	6	0.56	0.73	20.2	10.1	B	46.6 ± 0.9	
		FN-Q19-5	20	600	14	0.90	0.88	23.2	6.7	B*	36.8 ± 1.3	
FN-Q20	6/6	FN-Q20-1	20	540	12	0.64	0.84	19.7	6.2	A	32.1 ± 0.9	34.4 ± 0.6
		FN-Q20-2	20	600	14	0.94	0.85	34.4	9.9	B*	35.9 ± 0.8	
		FN-Q20-3	100	540	11	0.48	0.82	6.9	2.3	A	34.9 ± 2.0	
		FN-Q20-4	200	480	8	0.34	0.82	6.0	2.4	A	33.5 ± 1.6	
		FN-Q20-5	20	530	12	0.52	0.83	7.0	2.2	A	32.9 ± 2	
		FN-Q20-6	340	600	9	0.81	0.82	27.6	10.4	A*	34.8 ± 0.8	
FN-Q21	5/5	FN-Q21-1	100	470	9	0.41	0.86	17.6	6.6	A*	24.8 ± 0.5	24.6 ± 1.9
		FN-Q21-2	200	440	7	0.38	0.77	4.4	2.0	B*	20.9 ± 1.4	
		FN-Q21-3	380	510	5	0.40	0.70	6.4	3.7	A	22.3 ± 1	
		FN-Q21-4	300	510	7	0.50	0.75	7.4	3.3	A	20.5 ± 1	
		FN-Q21-5	300	510	7	0.52	0.76	9.4	4.2	B*	31.3 ± 1.3	
FN-Q22	3/3	FN-Q22-3	20	600	14	0.85	0.89	30.8	8.9	A	36.4 ± 0.9	35.8 ± 0.6
		FN-Q22-4	100	530	11	0.66	0.84	41.2	13.7	B	36.3 ± 0.5	
		FN-Q22-6	340	540	7	0.62	0.75	20.7	9.3	B	34.5 ± 0.8	
FN-Q23	0/1											
FN-Q24	3/4	FN-Q24-2	100	410	7	0.35	0.83	10.6	4.7	A*	34.7 ± 1.0	32.1 ± 2.6
		FN-Q24-3	20	340	6	0.32	0.77	17.5	8.8	A	28.7 ± 0.4	
		FN-Q24-5	20	470	10	0.58	0.87	9.4	3.3	A	37.3 ± 2.0	

[a] *n*/*N* shows successful versus attempted paleointensity determinations. T_{min} and T_{max} specify the temperature range of the straight line segment calculated over Np successive points. The fraction of the NRM (*f*), the gap factor (*g*) and the quality factor (*q*) were calculated according to Coe et al. (1978). *w* denotes the weighting factor of Prévot et al. (1985). C defines the class as described in the criteria section. *Indicates that check correction was applied. *H* and F_w are the paleointensity values with associated standard deviation (S.D.) for the individual determination and the weighted site mean, respectively.

Table 3
Mean directions for Fernando de Noronha[a]

	Declination (°)	Inclination (°)	k	α_{95} (°)	N	VGP_{long} (°)	VGP_{lat} (°)	dp	dm	S_f (°)	S_{low} (°)	S_{up} (°)
Quixaba formation												
All	0.6	−13.5	34	5.7	20	136.3	86.9	3.0	5.8	11.2	9.1	14.6
Normal	4.2	−10.3	32	7.5	13	75.5	85.6	3.8	7.5			
Reverse	173.7	19.1	101	6.0	7	13.6	−81.4	3.3	6.3			
Remedios formation												
All	357.6	−11.7	12	15.3	9	196.7	86.8	7.9	15.5	15.5	12.6	20.1
Normal	1.3	−17.3	23	16.5	5	133.2	84.8	8.8	17.0			
Reverse	172.8	4.1	8	35.7	4	71.4	−82.6	17.9	35.8			

[a]Shown are the mean directions and virtual geomagnetic poles for the directional groups of all, normal magnetized and reverse magnetized volcanic units of both formations. Additionally, the geomagnetic VGP dispersion corrected for the within-site scatter S_f (McElhinny and McFadden, 1997) with confidence limits S_{low} and S_{up} (Cox, 1969) is given for the overall mean of both formations. The intermediate direction of DG FN-RDG5, which shows an angle between GP latitude and geographic pole of more than 30° was excluded from calculation of mean values and VGP dispersion.

lava flows of this island formed 1.8 Myr ago during the Olduvai polarity interval. The mean direction for the Quixaba formation (Table 3) corresponds, within the margins of error, to an axial dipolar field. The geomagnetic VGP dispersion corrected for the within-site scatter S_f is slightly lower than expected for this latitude (see Table 3). Almost identical low values for S_f at equatorial sampling locations were also observed by Rochette et al. (1997). For calculation of the mean directions and VGP dispersion we used the directional groups of contiguous lava flows (Table 1). Directional groups in preference to single sites are used as long-term features of the geomagnetic field, recorded in volcanic units covering approximately 1.5 Ma, are the object of this examination. Such features may be heavily biased if numerous discrete directional results from volcanic units formed over a very short period of time outnumber other directional data. The application of directional groups relies upon the assumption that similar directions in successive lavas are exclusively related to volcanic events forming several flows within a very short time period. This supposition is thought to be justifiable, in cases wherein long-term features of the geomagnetic field are the object of investigation.

The age range and lithology of sampled volcanic units from the Remedios formation is more variable than for the Quixaba formation. Only two out of four sampled units could be analyzed from the oldest dated volcanic rocks on the island, the highly altered alkali basalts (sites FN-R14 to FN-R17) with an age of approximately 12.3 Myr (Cordani, 1970). The essexites (sites FN-R10 to FN-R12), with an age of 9.40 ± 0.25 Myr, show reverse directions and the alkali trachyte FN-R13 (10.8 Myr) is normally magnetized. Phonolitic intrusives of the Remedios section are abundant throughout the island and several sites were sampled. However, none of the sites yielding accurate mean directions is dated so far. Within the margins of error, the mean paleomagnetic direction of the Remedios formation (Table 3) is similar to the South American paleomagnetic pole for 10 Ma according to Besse and Courtillot (2002) and very similar to the mean results of the Quixaba formation.

Such a similarity between the mean directions of the Quixaba formation and the Remedios formation was not found by Richardson and Watkins (1967). Both the declination and inclination record of these authors show differences of more than 10° between the formations. In particular, the reverse directions of the Quixaba formation of Richardson and Watkins (1967) show a significant deviation from an axial dipolar field. These data were incorporated in the time-averaged field model of Johnson and Constable (1995) and added significant structure in the equatorial Atlantic. In Table 4, a comparison of the differences between an axial dipolar field and the observed declination and inclination values of the Quixaba formation between our study and the study of Richardson and Watkins (1967) are shown. Our results indicate that the directional anomaly of the reverse field is far smaller than observed during the previous study, but still present. The declination anomaly of the normal

Table 4
Average directional differences to an axial dipolar field[a]

	This study	Previous studies
Normal field		
ΔInclination	$-2.6^\circ \pm 3.8^\circ$ (13 DGs)	-0.9° (7 sites)
ΔDeclination	$4.2^\circ \pm 3.8^\circ$ (13 DGs)	-9.7° (7 sites)
VADM	$7.6 \pm 2.9 \times 10^{22}$ A m^2 (6 DGs)	–
Reverse field		
ΔInclination	$11.4^\circ \pm 3.0^\circ$ (7 DGs)	19.3° (5 sites)
ΔDeclination	$-6.3^\circ \pm 3.0^\circ$ (7 DGs)	-8.6° (5 sites)
VADM	$7.4 \pm 2.5 \times 10^{22}$ A m^2 (4 DGs)	–

[a] Observed paleodirections of the Quixaba formation were subtracted from the values which would be obtained at the sampling location in a purely axial dipolar field. Also shown is the virtual axial dipole moment (VADM) for both polarities.

field is smaller and has the opposite sign than previously observed. Furthermore, the reverse and normal mean directions are antipodal within the margin of errors. The differences shown in Table 4, however, are not related to the data analysis of individual sites as in all re-sampled lava flows of the Quixaba formation, our results are similar to those identified by Richardson and Watkins (1967) and Schult et al. (1986). In fact, the difference between the previous study and our results is linked to the extension of the data set in terms of time coverage and quantity of sites, and to the application of directional groups.

Although it is not possible to draw a global conclusion from the observations from one sampling location and, in particular, when only few independent directions and intensities are available, our results emphasize the importance of extending the paleomagnetic database, especially for regions presently represented by poor sampling coverage, in order to facilitate conclusive field models describing whether persistent long-term features of the geomagnetic field exist or not. For Fernando de Noronha our paleomagnetic results suggest that the present day dipole anomaly is not a persistent local feature of the Earth's magnetic field.

7.2. Paleointensity determination and alteration correction

Global coverage of absolute paleointensity data, when compared to that of paleodirectional results, is markedly deficient in data quantity and quality. Absolute paleointensities are difficult to obtain and require lengthy experimental procedures. Furthermore, the success rate of these experiments is generally poor, well below 50%. Rigorous pre-selection methods greatly enhance the success rate in case of our samples from Fernando de Noronha. The success rate commonly describes the ratio of obtained results versus attempted determinations, and not the ratio of obtained paleointensity versus obtained paleodirection of individual volcanic units. For our data the later ratio results in 25% and is comparable to other paleointensity studies. In addition to the rock magnetic pre-selection experiments, the correction for magnetomineralogical changes additionally increased the success rate. McClelland and Briden (1996) and Valet et al. (1996) investigated such correction methods and came to the same basic conclusion: alteration correction is valid, if all of the following conditions are fulfilled and can adequately demonstrated during laboratory treatment:

- The pTRM check, which is used for correcting the paleointensity determination, is sensible for alteration below the temperature at which the check is performed. Therefore, the alteration product, which formed during thermal demagnetization, must only affect unblocking temperatures below the subsequent alteration check.
- MD remanences and MD tails must not be present. The remaining NRM should not be affected during pTRM acquisition.
- There should be no ongoing alteration during repeated heating/cooling cycles to the same or lower temperatures.
- Checks must be performed successively and should cover the whole temperature spectrum of the experiment.

The modified paleointensity experiment, performed on the samples of Fernando de Noronha, satisfies most of these requirements. Alteration checks were performed continuously from 100 °C up to the maximum temperature after every second heating step. Additivity checks and repeated demagnetization steps are sensitive for MD remanence. Since two heating/cooling cycles are conducted between the pTRM check and the associated additivity check, ongoing alterations are also monitored by the additivity check. In addition to

MD tails, repeated demagnetization steps may be affected by destruction of remanence with blocking temperatures above the actual heating step. Formation of new remanence above the actual heating step is monitored, if it happens during a field-on step.

Alteration affecting higher blocking temperatures than the heating step of the alteration check and lower temperatures than the previous demagnetization step, however, cannot be monitored with any of the described checks during the experiment. This can cause an unrecognized failure to the paleointensity determination. The influence of this effect is reduced, however, if the temperature interval between check and demagnetization steps is small during the experiment. For all paleointensity experiments, where check correction is applied (Table 2), the maximum temperature difference was $\leq 60\,^{\circ}C$ at heating steps above $380\,^{\circ}C$. The heating step dependent ARM AF demagnetization experiments (see Fig. 5), show no significant alteration at temperatures below $400\,^{\circ}C$ and only slight alteration above $400\,^{\circ}C$ for samples where the check correction was applied. This test is sensitive to all alterations during laboratory thermal treatment. However, due to the different grain size dependence of ARM and TRM (e.g. Yu et al., 2003), the ARM test can only be a first order approximation of alteration occurring during Thellier-type experiments. For seven specimens from the collection, a paleointensity determination would have not been possible without check correction (e.g. Fig. 7b). The majority of the other specimens, where check correction is applied, show results similar to those obtained when non-corrected, within a smaller fraction of the NRM (Fig. 7a). Using only non-corrected determinations would result in very similar mean paleointensity values for most sites, albeit with higher within-site scatter. This finding is a further indication that alteration correction improves the results. When comparing non-corrected and corrected paleointensity determinations in one site, similar results for most lavas, except sites FN-Q16 and FN-Q15, are found. None of the specimens of site FN-Q16 could be analyzed without check correction, but all of the applied checks during the determination and thermal dependent ARM demagnetizations on sister samples point to a suitability for analysis, resulting in similar values for the individual specimens. Site FN-Q15 shows high within-site scatter, whether or not applying check correction.

In addition to the directional differences from an axial dipolar field, the virtual axial dipole moments were calculated from the mean paleointensity values of normal and reverse geomagnetic field states (Table 4). The values are similar for the reverse and normal state of the field and the overall mean value of $(7.5 \pm 2.6) \times 10^{22}\,A\,m^2$ is only slightly lower than the present day magnetic moment. This observation is in agreement with the findings of Heller et al. (2002) whose paleointensity database analysis points to a Pliocene dipole moment close to $8 \times 10^{22}\,A\,m^2$. These authors suggest that submarine basaltic glass (SBG) samples, which are included in the paleointensity database analysis of Juarez and Tauxe (2000), are biased by chemical alteration processes, leading to an underestimation of the paleointensity. Smirnov and Tarduno (2003) came to a similar conclusion, suggesting systematically too low paleointensities for SBG older than 0.5 Ma. Juarez and Tauxe (2000) obtained an average value of $5.5 \times 10^{22}\,A\,m^2$ for the last 5 Myr, significantly lower than the mean value obtained for Fernando de Noronha. Hence, our study phenomenologically supports the argumentation of Heller et al. (2002) and Smirnov and Tarduno (2003).

8. Conclusions

(1) Pre-selection of samples using a combination of rock magnetic experiments, ore microscopy, and ARM demagnetizations after different heating steps, greatly enhanced the success rate of paleointensity measurements for our samples from Fernando de Noronha.
(2) Additivity checks monitor the influence of MD remanence during Thellier-type experiments and ongoing alteration during repeated heating steps. Furthermore, these checks provide a measure for the quality of corrections for magnetomineralogical changes. The application of alteration correction further enhanced the success rate and leaded to a reduction of the within-site scatter of paleointensity determination indicating an improvement of the results.
(3) The mean paleomagnetic poles for the Quixaba and Remedios formation are similar to those expected from an axial dipolar field. Analyses of paleomagnetic mean values show that the

geomagnetic dipole anomaly during the pliocene is far smaller in this region than previously found in paleomagnetic data and that observed for the present day field.

(4) Similar VADMs are observed for normal and reverse field states during the Pliocene on Fernando de Noronha. The obtained values are similar to the present day magnetic moment.

Acknowledgements

We would like to thank the Instituto Brasileiro do Meio Ambiente (IBAMA) for sampling permissions and their support during the field work. Axel Schult, Heiner Soffel, Nikolai Petersen, David Krása, Christoph Heunemann, Maria Antretter and Kari Anderson are acknowledged for many useful discussions. We thank the journal reviewers for helpful and constructive comments. Sampling was funded by the German Science Foundation (So72/67-3).

References

Almeida, F.F.M., 1958. Geologia e petrologia do archipelago de Fernando de Noronha. Divisao de Geologia e Mineralogia 13.

Besse, J., Courtillot, V., 2002. Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr. J. Geophys. Res. 107, 2300. 10.1029/2000JB000050.

Butler, R.F., 1992. Paleomagnetism: Magnetic Domains to Geologic Terranes. Blackwell Scientific Publications, Oxford.

Carlut, J., Courtillot, V., 1998. How complex is the time-averaged geomagnetic field over the past 5 Myr? Geophys. J. Int. 134, 527–544.

Coe, R.S., 1967. Paleointensity of the Earth's magnetic field determined from tertiary and quaternary rocks. J. Geophys. Res. 72, 3247–3262.

Coe, R.S., Grommé, S., Mankinen, E.A., 1978. Geomagnetic paleointensities from radiocarbon-dated lava flows on Hawaii and the question of the Pacific nondipole low. J. Geophys. Res. 83, 1740–1756.

Cordani, U.G., 1970. Idade do vulcanismo no Oceano Atlantico sul. Bol. IGA 1, 9–75.

Cox, A., 1969. Confidence limits for the precision parameter *k*. Geophys. J. Roy. Astr. Soc. 18, 545–549.

Day, R., Fuller, M.D., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain size and composition dependence. Phys. Earth Planet. Inter. 13, 260–266.

Dunlop, D.J., Özdemir, O., 2000. Effect of grain size and domain state on thermal demagnetization tails. Geophys. Res. Lett. 27, 1311–1314.

Fabian, K., 2001. A theoretical treatment of paleointensity determination experiments on rocks containing pseudo-single or multi domain magnetic particles. Earth Planet. Sci. Lett. 188, 45–58.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. Roy. Soc. London A 217, 295–305.

Gubbins, D., Kelly, P., 1993. Persistent patterns in the geomagnetic field over the past 2.5 Myr. Nature 365, 829–832.

Hatakeyama, T., Kono, M., 2002. Geomagnetic field model for the last 5 Myr: time-averaged field and secular variation. Phys. Earth Planet. Inter. 133, 181–215.

Heller, R., Merrill, R.T., McFadden, P.L., 2002. The variation of intensity of the earth's magnetic field with time. Phys. Earth Planet. Inter. 131, 237–249.

Johnson, C.L., Constable, C.G., 1995. The time-averaged geomagnetic field as recorded by lava flows over the past 5 Myr. Geophys. J. Int. 122, 489–519.

Johnson, C.L., Constable, C.G., 1997. The time-averaged geomagnetic field: global and regional biases for 0–5 Ma. Geophys. J. Int. 131, 643–666.

Juarez, M.T., Tauxe, L., 2000. The intensity of the time-averaged geomagnetic field: the last 5 Myr. Earth Planet. Sci. Lett. 175, 169–180.

Kono, M., Hiroi, O., 1996. Paleosecular variation of field intensities and dipole moments. Earth Planet. Sci. Lett. 139, 251–262.

Kosterov, A.A., Prévot, M., 1998. Possible mechanisms causing failure of Thellier paleointensity experiments in some basalts. Geophys. J. Int. 134, 554–572.

Krása, D., Heunemann, C., Leonhardt, R., Petersen, N., 2003. Experimental procedure to detect multidomain remanence during Thellier–Thellier experiments. Phys. Chem. Earth 28, 681–687.

Matzka, J., Krása, D., Kunzmann, T., Schult, A., Petersen, N., 2003. Magntic state of 10–40 Ma old ocean basalts and its implication for natural remanent magnetization. Earth Planet. Sci. Lett. 206, 541–553.

Mazoires, J., Menor, E.A., Roche, A., Saucier, H., Saucier, M., Sial, A.N., 1971. Magnétisme terrestre—etude paléomagnétique de lavas de l'ile fernando de noronha. CR Acad. Sci. Paris 273, 551–554.

McClelland, E., Briden, J.C., 1996. An improved methodology for Thellier-type paleointensity determination in igneous rocks and its usefulness for verifying primary thermoremanence. J. Geophys. Res. 101, 21995–22013.

McElhinny, M.W., McFadden, P.L., 1997. Paleosecular variation over the past 5 Myr based on a new generalized database. Geophys. J. Int. 131, 240–252.

McElhinny, M.W., McFadden, P.L., Merrill, R.T., 1996. The myth of the Pacific dipole window. Earth Planet. Sci. Lett. 143, 13–22.

McFadden, P.L., McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in paleomagnetism. Earth Planet. Sci. Lett. 87, 161–172.

Prévot, M., Mankinen, E.A., Coe, R.S., Grommé, S., 1985. The Steens Mountain (Oregon) geomagnetic polarity transition. 2. Field intensity variations and discussion of reversal models. J. Geophys. Res. 90, 10417–10448.

Richardson, A., Watkins, N.D., 1967. Paleomagnetism of Atlantic islands: Fernando Noroñha. Nature 215, 1470–1473.

Riisager, P., Riisager, J., 2001. Detecting multidomain magnetic grains in Thellier paleointensity experiments. Phys. Earth Planet. Inter. 125, 111–117.

Rivalenti, G., Mazzucchelli, M., Girardi, V.A.V., Vannucci, R., Adelaide Barbieri, M., Zanetti, A., Goldstein, S.L., 2000. Composition and processes of the mantle lithosphere in northeastern Brazil and Fernando de Noronha: evidence from mantle xenoliths. Contr. Miner. Petrol. 138, 308–325.

Rochette, P., Ben Atig, F., Collombat, H., Vandamme, D., Vlag, P., 1997. Low paleosecular variation at the equator: a paleomagnetic pilgrimage from Galapagos to Esterel with Allan Cox and Hans Zijderveld. Geologie Mijnbouw 76, 9–19.

Schult, A., Calvo Rathert, M., Guerreiro, S., Bloch, W., 1986. Paleomagnetism and rock magnetism of Fernando de Noronha. Brazil. Earth Planet. Sci. Lett. 79, 208–216.

Selkin, P.A., Tauxe, L., 2000. Long-term variations in paleointensity. Phil. Trans. Roy. Soc. Lond. 358, 1065–1088.

Shaw, J., 1974. A new method of determining the magnitude of the paleomagnetic field: application to five historic lavas and five archaeological samples. Geophys. J. Roy. Astr. Soc. 39, 133–141.

Smirnov, A.V., Tarduno, J.A., 2003. Magnetic hysteresis monitoring of Cretaceous submarine basaltic glass during Thellier paleointensity experiments: evidence for alteration and attendant low eld bias. Earth Planet. Sci. Lett. 206, 571–585.

Ulbrich, M.N.C., Maringolo, V., Ruberti, E., 1994. The geochemistry of alkaline volcanic–subvolcanic rocks from the Brazilian Fernando de Noronha archipelago, Southern Atlantic Ocean. Geochim. Brasil. 8, 21–39.

Valet, J.-P., Brassart, J., Le Meur, I., Soler, V., Quidelleur, X., Tric, E., Gillot, P.-Y., 1996. Absolute paleointensity and magneto-mineralogical changes. J. Geophys. Res. 101 (B11), 25029–25044.

Wehland, F., Leonhardt, R., Appel, E., submitted. Magnetic interaction analysis of basaltic samples. Geophys. J. Int.

Yu, Y., Dunlop, D.J., Özdemir, O., 2003. Are ARM and TRM analogs? Thellier analysis of ARM and pseudo-Thellier analysis of TRM. Earth Planet. Sci. Lett. 205, 325–336.